

## 2.3 Combinations

A combination is a selection of objects from a group where the order is not relevant. An unordered selection of  $r$  objects from a set of  $n$  objects is called a combination.

For example, if a committee is being selected and there will be a president, vice president, and treasurer, then order matters so it is a permutation problem. But if a committee of three people is being formed without specific roles, then order does not matter and it is a combination problem.

The number of selections of  $n$  distinct objects taking  $r$  at a time is the combination denoted by  $nC_r$  or  $C(n, r)$

$$nC_r = \frac{n!}{r!(n-r)!}$$

In particular,  $nC_n = 1$ ,  $nC_{n-1} = n$ ,  $nC_{n-r} = nC_r$

### Worked Example. 2.3

#### Example 2.3.1

Find the number of arrangements of the letter in the word TALLAHASSEE. How many arrangements have no adjacent A's?

**Solution:**

There are 11 letters in the word TALLAHASSEE. The letter 'A' is repeating 3 times, 'L' repeating 2 times, 'S' is repeating 2 times and 'E' is repeating 2 times in the word.

Number of arrangements of the letter =  $\frac{11!}{3! \times 2! \times 2! \times 2!} = 831,600$  ways

If we remove the A's, the letters become TLLHSSEE.

Number of arrangements of the word without A's =  $\frac{8!}{2! \times 2! \times 2!} = 5040$  ways.

We can place A in the following 9 places

$\downarrow$  T  $\downarrow$  L  $\downarrow$  L  $\downarrow$  H  $\downarrow$  S  $\downarrow$  S  $\downarrow$  E  $\downarrow$  E  $\downarrow$   
 A    A    A    A    A    A    A    A    A

Three of these locations can be selected in  ${}^9C_3 = 84$  ways, and this is also possible for all the other 5039 arrangements. By the rule of product there are  $5040 \times 84 = 423,360$  arrangements of the letter in TALLAHASSEE with no consecutive A's.

**Example 2.3.2**

A student is to answer seven out of 10 questions on an examination. In how many ways can he make his selection if (a) there are no restrictions? (b) he must answer the first two questions? (c) he must answer at least four of the first six questions?

**Solution:**

- a) Number of ways of selection of 7 questions out of 10 =  ${}^{10}C_7 = 120$ .
- b) After selecting first two questions, he must select 5 questions from the remaining 8 questions. This can be done in  ${}^8C_5 = 56$  ways.
- c) He must answer four of the first 6 questions and 3 from the remaining 4 questions in  $({}^6C_4 \times {}^4C_3)$  ways, four of the first 5 questions and 2 from the remaining 4 questions in  $({}^6C_5 \times {}^4C_2)$  ways and six of the first 6 questions and 1 from the remaining 4 questions in  $({}^6C_6 \times {}^4C_1)$  ways. Therefore, number of ways of selecting at least four of the first six questions is  ${}^6C_4 \times {}^4C_3 + {}^6C_5 \times {}^4C_2 + {}^6C_6 \times {}^4C_1 = 100$  ways.

**Example 2.3.3**

A committee of eight is to be formed from 16 men and 10 women. In how many ways can the committee be formed if (a) there are no restrictions (b) there must be 4 men and 4 women (c) there should be an even number of women (d) more women than men (e) at least 5 men.

**Solution:**

(a) If there are no restrictions between men and women, the number of ways 8 persons are chosen from 26 persons is  ${}^{26}C_8 = \frac{26!}{8! \times 18!} = 1,562,275$ .

(b) Choose 4 men out of 16 men in  $({}^{16}C_4)$  ways and choose 4 women out of 10 women  $({}^{10}C_4)$  ways. The number of ways the committee be formed having 4 men and 4 women is  ${}^{16}C_4 \times {}^{10}C_4 = 382,200$ .

(c) Select 2 women from 10 women and 6 men from 16 men in  $({}^{10}C_2 \times {}^{16}C_6)$  ways, select 4 women from 10 women and 4 men from 16 men in  $({}^{10}C_4 \times {}^{16}C_4)$  ways, select 6 women from 10 women and 2 men from 16 men in  $({}^{10}C_6 \times {}^{16}C_2)$  ways and select 8 women from 10 women and 0 men from 16 men  $({}^{10}C_8 \times {}^{16}C_0)$

Therefore, the number of ways of selecting even number of women is  ${}^{10}C_2 \times {}^{16}C_6 + {}^{10}C_4 \times {}^{16}C_4 + {}^{10}C_6 \times {}^{16}C_2 + {}^{10}C_8 \times {}^{16}C_0 = 767,805$ .

(d) Select 5 women from 10 women and 3 men from 16 men in  $({}^{10}C_5 \times {}^{16}C_3)$  ways, select 6 women from 10 women and 2 men from 16 men in  $({}^{10}C_6 \times {}^{16}C_2)$  ways, select 7 women from 10 women and 1 men from 16 men in  $({}^{10}C_7 \times {}^{16}C_1)$  ways and select 8 women from 10 women and 0 men from 16 men in  $({}^{10}C_8 \times {}^{16}C_0)$  ways.

Therefore, the required number of ways =  ${}^{10}C_5 \times {}^{16}C_3 + {}^{10}C_6 \times {}^{16}C_2 + {}^{10}C_7 \times {}^{16}C_1 + {}^{10}C_8 \times {}^{16}C_0 = 168,285$

(e) Select 6 men from 16 men and 2 women from 10 women in  $({}^{16}C_6 \times {}^{10}C_2)$  ways, select 7 men from 16 men and 1 women from 10 women in  $({}^{16}C_7 \times {}^{10}C_1)$  ways and select 8 men from 16 men and 0 women from 10 women in  $({}^{16}C_8 \times {}^{10}C_0)$  ways.

(f) Therefore, the required number of ways is  ${}^{16}C_6 \times {}^{10}C_2 + {}^{16}C_7 \times {}^{10}C_1 + {}^{16}C_8 \times {}^{10}C_0 = 487,630$ .

**Example 2.3.4**

An urn contain 15 balls 8 of which are red and 7 are black. In how many ways can 5 balls be chosen so that a) all five are red b) ) all five are black c) 2 are red and 3 are black d) 3 are red and 2 are black.

**Solution:**

a) Five red balls can be selected from 8 red balls  ${}^8C_5 = 56$  ways.

b) Five black balls can be selected from 7 black balls  ${}^7C_5 = 21$  ways.

c) Two red balls can be selected from 8 red balls  ${}^8C_2 = 28$  ways and three black balls can



be selected from 7 black balls in  ${}^7C_3 = 35$  ways. By the rule of product, the selection can be made in  $28 \times 35 = 980$  ways.

d) Three red balls can be selected from 8 red balls  ${}^8C_3 = 56$  ways and two black balls can be selected from 7 black balls in  ${}^7C_2 = 21$  ways. By the rule of product the selection can be made in  $56 \times 21 = 1176$  ways.

### Example 2.3.5

A choir director must select six hymns for a Sunday church service. She has three hymn books, each containing 25 hymns (there are 75 different hymns in all). In how many ways can she select the hymns if she wishes to select (a) two hymns from each book? (b) at least one hymn from each book?

**Solution:**

a) Number of ways of selecting 2 hymns from the each book is  ${}^{25}C_2 \times {}^{25}C_2 \times {}^{25}C_2 = ({}^{25}C_2)^3$  ways.

b) Number of ways of selecting four hymns from one book, one from each of the other two books is  $3 \times ({}^{25}C_4 \times {}^{25}C_1 \times {}^{25}C_1)$  ways, the number of ways of selecting one hymns from one book, two from the other book and 3 from the next book is  $6 \times ({}^{25}C_1 \times {}^{25}C_2 \times {}^{25}C_3)$  ways and the number of ways of selecting two hymns from each book is  $({}^{25}C_2)^3$ .

By the rule of sum the number of ways of selecting at least one book is  $3 \times ({}^{25}C_4 \times {}^{25}C_1 \times {}^{25}C_1) + 6 \times ({}^{25}C_1 \times {}^{25}C_2 \times {}^{25}C_3) + ({}^{25}C_2)^3$ .

### Example 2.3.6

How many arrangements of the letter MISSISSIPPI have no consecutive S's?

**Solution:**

If we remove all S, the remaining letters, M, I, I, I, P, P, I, can be arranged in  $\frac{7!}{4! \times 2!} = 105$  ways, since I occupies 4 places and 'P' occupies 2 places.

We can place S in the following 8 places

↓ M ↓ I ↓ I ↓ I ↓ P ↓ P ↓ I ↓  
S S S S S S S S

Four of these locations can be selected in  ${}^8C_4 = 70$  ways. By the rule of product the number of arrangements with no adjacent S's =  $105 \times 70 = 7350$  ways.

**Example 2.3.7**

In how many ways can 12 different books be distributed among four children so that a) each child gets three books? b) the two oldest children get four books each and the two youngest get two books each?

**Solution:**

12 different books be distributed among four children in  $12C_4 = 495$  ways.

a) The first 3 books can be selected in  $12C_3$  different ways for the first child. The next 3 books can be selected in  $9C_3$  ways for the second child. The third child will get 3 books in  $6C_3$  ways and the fourth child will get 3 books set in  $3C_3$  ways. So the 12 books can distributed among four children in  $12C_3 \times 9C_3 \times 6C_3 \times 3C_3 = 369,600$  ways.

b) The first oldest child will get four books in  $12C_4$  ways, Second oldest child will get four books in  $8C_4$  ways, the third youngest child will get two books in  $4C_2$  ways and the fourth youngest child will get two books in  $2C_2$  ways. So the total distribution is  $12C_4 \times 8C_4 \times 4C_2 \times 2C_2 = 207,900$  ways.

**Example 2.3.8**

How many ways are there to pick a five-person basketball team from 12 possible players? How many selections include the weakest and the strongest players?

**Solution:**

We can select 5 persons from 12 players in  $12C_5 = 792$  different ways. First, we select weakest and the strongest players from 12 players. After fixing these two players remaining 3 players can be selected from 10 players in  $10C_3 = 120$  ways

**Example 2.3.9**

In the manufacture of a certain type of automobile, four kinds of major defects and seven kinds of minor defects can occur. For those situations in which defects do occur, in how many ways can there be twice as many minor defects as there are major ones?

**Solution:**

From four kinds of major defects and seven kinds of minor defects, we can select twice as many minor defects as major defects in  $4C_1 \times 7C_2 + 4C_2 \times 7C_4 + 4C_3 \times 7C_6 = 322$  ways.

**Example 2.3.10**

A physical education teacher must make up 4 volley ball teams A,B,C and D of nine girls each from the 36 girls in the class. In how many ways can she select these 4 teams?

**Solution:**

To form the team A, the teacher can select any 9 girls from the 36 girls in  ${}^{36}C_9$  ways. From the remaining 27 girls the teacher can form the team B in  ${}^{27}C_9$  ways. This leaves  ${}^{18}C_9$  and  ${}^9C_9$  possible ways to select teams C and D respectively.

By the rule of product, the four teams can be chosen in  ${}^{36}C_9 \times {}^{27}C_9 \times {}^{18}C_9 \times {}^9C_9 = 2145 \times 10$  ways.

**Exercise 2.3**

1. A committee of 12 is to be selected from 10 men and 10 women. In how many ways can the selection be carried out if a) there is no restriction b) there must be 6 men and 6 women c) there must be even number of women d) there must be more than women than men e) there must be at least 8 men.
2. How many ways are there to pick a five-person basketball team from 12 possible players? How many selections include the weakest and the strongest?
3. A gym coach must select 11 seniors to play on a football team. If he can make his selection in 12,367 ways, how many seniors are eligible to play?
4. In how many ways can the letters in WONDERING be arranged with exactly two consecutive vowels?
5. A committee of 4 is to be selected from amongst 5 boys and 6 girls. In how many ways can this be done so as to include a) exactly one girl b) at least one girl.
6. A person has got 12 friends of whom 8 are relatives. In how many ways can he invite 7 guests such that 5 of them may be relatives?
7. Out of 3 books on Mathematics, 4 books on Physics and 5 books on Chemistry, how many collections can be made, if each collection consists of a) exactly one book on each subject b) at least one book on each subject?
8. In how many ways can the letters in ASSASSINATION be arranged such that (a) there is no restriction (b) all the three S's together? c) No consecutive S's.
9. In how many ways can a teacher distribute 12 different science books among 16 students if a) no student gets more than one book? b) the oldest student gets two books but no other student gets more than one book?

**Answers:**

1. a)  ${}^{20}C_{12}$     b)  ${}^{10}C_6 \cdot {}^{10}C_6$     c)  ${}^{10}C_2 \cdot {}^{10}C_{10} + {}^{10}C_4 \cdot {}^{10}C_8 + {}^{10}C_6 \cdot {}^{10}C_6 + {}^{10}C_8 \cdot {}^{10}C_4$   
d)  ${}^{10}C_7 \cdot {}^{10}C_5 + {}^{10}C_8 \cdot {}^{10}C_4 + {}^{10}C_9 \cdot {}^{10}C_3 + {}^{10}C_{10} \cdot {}^{10}C_2$   
e)  ${}^{10}C_4 \cdot {}^{10}C_8 + {}^{10}C_3 \cdot {}^{10}C_9 + {}^{10}C_2 \cdot {}^{10}C_{10}$



$$2. 12C_5, 10C_3$$

$$3. nC_{11} = 12367, \quad n = 17. \quad 4. \left(\frac{7!}{2!}\right) 6^3$$

$$5. a) 60 \quad b) 325$$

$$6. 8C_5 \cdot 4C_2$$

$$7. a) 60 \quad b) 3255$$

$$8. 360$$

$$9. a) \frac{16!}{4!} \quad b) 12C_2 \frac{15!}{5!}$$